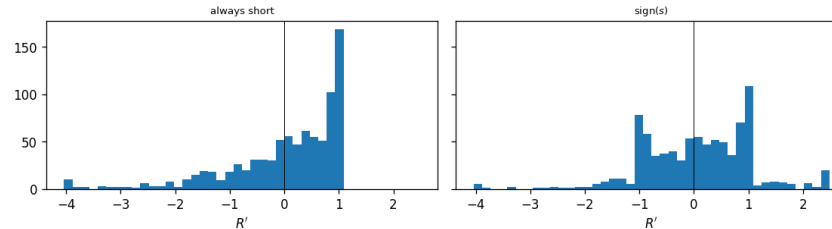


0DTE nearest-OTM straddle: per-rule return summary, entry 15:30 ET, held to cash-settle, models compared on the same 866 days

	n	mean	std	min	max	skew	ex. kurt	t	Sharpe _{clk}	n_{buy}	%buy	Sharpe _{clk} ^x
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.017	1.115	-9.38	1.08	-2.26	9.18	0.44	0.24	0	0.0	-0.23
<i>sign(s) (± 1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.064	1.114	-6.37	9.38	0.12	9.07	1.70	0.92	286	33.0	0.44
block-diagonal ridge	866	0.091	1.112	-6.37	9.38	0.62	8.95	2.41	1.30	346	40.0	0.83
block-diagonal ridge, without the FOMC columns	866	0.092	1.112	-4.45	9.38	1.01	8.82	2.45	1.32	355	41.0	0.85
LightGBM [†]	866	0.104	1.111	-6.37	9.38	0.62	8.96	2.76	1.49	290	33.5	1.01
XGBoost [†]	866	0.102	1.111	-6.37	9.38	0.52	9.00	2.72	1.46	279	32.2	0.99
lasso (causally tuned) [†]	866	0.103	1.111	-6.37	9.38	0.40	9.04	2.74	1.48	321	37.1	1.00
lasso ($\alpha = 10^{-4}$)	866	0.105	1.111	-6.37	9.38	0.95	8.84	2.79	1.51	358	41.3	1.03
elastic net (causally tuned) [†]	866	0.075	1.113	-6.37	9.38	0.27	9.05	1.98	1.07	307	35.5	0.60

block-diagonal ridge, midpoint fills: entry 15:30 ET, held to cash-settle; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



Distribution of the return of the position taken at 15:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window; the mass at -1 is the days the straddle expires worthless).

One trade a day: enter the nearest-OTM straddle at 15:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread). **Straddle** here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean}/\text{std}$, with no autocorrelation correction. “ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of that clock’s series: one remaining-session short per expiration day (enter at this clock, hold those strikes to the close, delta-hedge every 30 minutes). It answers “if I only entered at this clock, what is my annual Sharpe?” It is *not* the book’s $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of $R_d^{\text{day}} = \sum_t R'_{d,t}$, the sum of the day’s thirteen overlapping remaining-session books. That reserved name is the book’s daily P&L after pooling the session. The thirteen books overlap; they are not thirteen successive holds. Sharpe^x is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the thirteen positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook’s column. The other column this table adds to the deck’s is $\text{Sharpe}_{\text{ann}}^x$, the same rule’s annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck’s.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). [†] marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - IV_{\text{hr}}^2 h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day’s remaining-session share of realized variance, $RV_t / \sum_{s \geq t} RV_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck’s treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat. At the 09:35 entry the forecast is the one issued at 09:30 for the 09:30–10:00 bar, w_t is that bar’s share, and h_t is the chain’s hours to the close.